WORKED SOLUTION

STEP II 2022, Question 9

Mechanics • Friction and Moments

STEP II 2022, Question 9

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

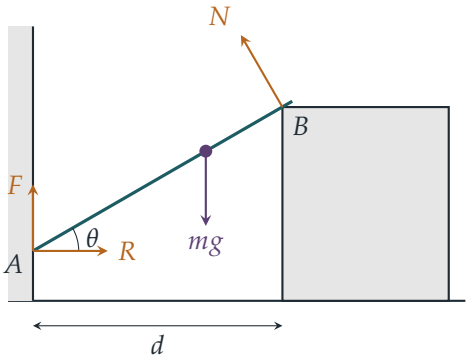
The plank is light, so only three things push on it: the point mass, the reaction at the prism's edge, and the wall. Taking moments about the point where the plank touches the wall removes both wall forces at once and gives the reaction at the edge straight away. Resolving vertically then gives the friction, and every later part is the friction inequality $|F| \leq \mu R$, rearranged.

Solution

Model. The plank is a light rigid rod, the mass is a particle of weight mg, the edge of the prism is smooth, and the wall is rough with coefficient μ. Let A be the point where the plank touches the wall and B the point where it rests on the edge. Since B is a horizontal distance d from the wall,

$$AB = d\sec\theta.$$

At B the plank rests on an edge, so the smooth reaction N is perpendicular to the *plank*. At A the wall gives a horizontal normal reaction R and a vertical friction force F, which we take as positive upwards (a negative F means it acts downwards).



Moments about A. Both wall forces pass through A. The weight acts at distance x along the plank, so its moment arm is $x\cos\theta$; N is perpendicular to the plank at distance $d\sec\theta$:

$$N\,d\sec\theta = mg\,x\cos\theta \quad\Longrightarrow\quad N = \frac{mg\,x\cos^2\theta}{d}.$$

Resolving. The direction of N makes an angle θ with the vertical, tilted towards the wall. Horizontally and vertically,

$$R = N\sin\theta = \frac{mg\,x\cos^2\theta\sin\theta}{d},$$
$$F = mg - N\cos\theta = mg\left(1 - \frac{x\cos^3\theta}{d}\right).$$

Note that $R > 0$, so the plank really does press on the wall.

Part (i)

If $x = d\sec^3\theta$ then $x\cos^3\theta/d = 1$, so $F = 0$: no friction acts at the wall.

Part (ii)

Equilibrium with friction requires $|F| \le \mu R$.

If $x > d\sec^3\theta$, then $x\cos^3\theta/d > 1$, so $F < 0$ (friction acts downwards; without it the end A would slide up the wall). The condition $-F \le \mu R$ reads

$$mg\left(\frac{x\cos^3\theta}{d} - 1\right) \le \mu\,\frac{mg\,x\cos^2\theta\sin\theta}{d}.$$

Multiplying by the positive quantity $d\sec^3\theta/(mg)$ gives

$$x - d\sec^3\theta \le \mu\,x\tan\theta, \qquad \text{so} \qquad \mu \ge \frac{x - d\sec^3\theta}{x\tan\theta},$$

dividing by $x\tan\theta > 0$ (as θ is acute). This is the required result.

If $x < d\sec^3\theta$, then $F > 0$ and the condition $F \le \mu R$ becomes, in the same way,

$$\mu \ge \frac{d\sec^3\theta - x}{x\tan\theta}.$$

> **Ask yourself** Which way does friction act, and how do we know without guessing?

We don't need to guess. Taking F positive upwards, the vertical equation tells us its sign. Friction can always be written as $|F| \le \mu R$, and the sign of F just decides which of the two inequalities we get.

Part (iii)

Both cases of (ii) say $|x - d\sec^3\theta| \le \mu x \tan\theta$, which is the same as the pair

$$x - d\sec^3\theta \le \mu x\tan\theta \qquad \text{and} \qquad d\sec^3\theta - x \le \mu x\tan\theta.$$

The second gives $d\sec^3\theta \le x(1+\mu\tan\theta)$. Since $1+\mu\tan\theta > 0$,

$$\frac{x}{d} \ge \frac{\sec^3\theta}{1+\mu\tan\theta}.$$

The first gives $x(1-\mu\tan\theta) \le d\sec^3\theta$. If $\mu < \cot\theta$ then $\mu\tan\theta < 1$, so $1-\mu\tan\theta > 0$ and we may divide without reversing the inequality:

$$\frac{x}{d} \le \frac{\sec^3\theta}{1-\mu\tan\theta}.$$

(If $\mu \ge \cot\theta$ there is no upper bound: the friction can hold the mass however far up the plank it sits.)

Part (iv)

The mass lies between the wall and the edge exactly when $x < AB = d\sec\theta$. Combining with the lower bound from (iii),

$$\frac{d\sec^3\theta}{1+\mu\tan\theta} \le x < d\sec\theta.$$

Dividing by $d\sec\theta > 0$ and multiplying by $1+\mu\tan\theta > 0$,

$$\sec^2\theta < 1+\mu\tan\theta \quad\Longleftrightarrow\quad \tan^2\theta < \mu\tan\theta,$$

using $\sec^2\theta = 1+\tan^2\theta$. Since $\tan\theta > 0$ we can divide by it, giving $\tan\theta < \mu$, as required.

Checks

- Take $\theta = 45^\circ$, $d = 1$ and $\mu = 0.5$. Then $\sec^3\theta = 2\sqrt{2} \approx 2.83$, and (iii) allows $1.89 \le x \le 5.66$. At $x = 2$, say, $F = mg(1 - 2/2\sqrt{2}) \approx 0.29mg$ and $N = mg$, $R = mg/\sqrt{2}$ and $\mu R \approx 0.35mg$, so equilibrium holds. Here the mass is beyond the edge ($AB = \sqrt{2}$), consistent with (iv), since $\tan 45^\circ = 1 > \mu$.
- Limiting case $\mu = 0$ (smooth wall): the bounds in (iii) collapse to $x = d\sec^3\theta$, the only position from (i).
- A computer check over many random values of θ, d, μ and x confirmed that the forces balance (including moments about other points), and that $|F| \le \mu R$ holds exactly when the inequalities of (iii) hold.

Common slips

- Taking the reaction at the edge as vertical. The plank rests on an edge, so the smooth reaction is perpendicular to the plank, not to the top of the prism.
- Using d instead of $d\sec\theta$ for the distance AB in the moments equation.
- Dividing $x(1-\mu\tan\theta) \leq d\sec^3\theta$ without checking the sign of $1-\mu\tan\theta$; that is why the condition $\mu < \cot\theta$ appears.

Beyond STEP

This plank is *statically determinate*: three unknown reactions and three equilibrium equations, so friction is pinned down exactly. Add one more support and the equations no longer fix the forces; engineers then bring in the elasticity of the beam to decide how the load is shared. The bounds in (iii) are a small example of a "region of equilibrium", a theme that returns in the study of stability. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body statics and friction carefully.

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